

Bloomberg Data Scientist Case Study

Predicting Arrival Delays in 2024 U.S. Flight Operations

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1. Executive Summary

This case study analyzes 2024 U.S. flight operations from the [BTS Reporting Carrier On-Time Performance dataset](#) (~7 million completed flights) to understand drivers of arrival delays and build a pre-flight model to predict whether a flight will arrive 15 or more minutes late.

Three core findings:

- **Delay risk is highly time-dependent.** Delay rates rise steadily from ~9% in the early morning to ~30% in the evening (19:00–20:00 departure), and July is the highest-delay month at 30%.
- **Carrier, airport, and route differences are substantial.** After filtering to meaningful volume, top-decile carriers and origin airports show delay rates 30–40% higher than the annual average, supporting their use as predictive features.
- **A tuned XGBoost model achieves the best operating point.** On the October–December holdout, the model reaches ROC-AUC 0.62 with the threshold tuned to 0.20, catching 67% of delayed flights at a precision of ~21%. This is a reasonable tradeoff for a broad delay-screening use case where recall is prioritized, though the final operating threshold should depend on the cost of false alarms.

The work is structured around a strict pre-flight modeling constraint: only fields available before scheduled departure are used as predictors. Actual departure time, delay-cause fields, and arrival fields are reserved for EDA and a potential post-departure model, not the pre-flight model.

2. Approach

2.1 Data Preparation

The dataset consists of twelve monthly CSV files from BTS covering January through December 2024, filtered to completed flights (cancelled and diverted flights excluded since they have no arrival-delay outcome), and consolidated into a single dataset of 6.97M records.

Memory-conscious processing was important given the data scale:

- **Selective column reads:** only the 36 columns relevant for EDA and modeling.
- **Type optimization:** downcasting integers and floats, converting high-cardinality strings to categoricals, reducing memory from several GB to ~800MB.
- **Sampling for modeling:** stratified sample of 500,000 flights for efficient model comparison while preserving target rate and key strata (month, hour).

2.2 Feature Engineering

All features were designed to be available before scheduled departure to avoid leakage:

- **Calendar:** Month, DayOfWeek, day_name, week, is_weekend, is_holiday, is_holiday_week (± 2 days of federal holiday).
- **Schedule timing:** sched_dep_hour, sched_arr_hour, CRSDepTime, CRSArrTime, CRSElapsedTime.
- **Network:** Reporting_Airline, Origin, Dest, route (Origin-Dest concatenation), Distance, DistanceGroup.
- **Operational scale:** origin_annual_volume, dest_annual_volume as airport-size proxies.

Fields explicitly excluded from the pre-flight model to prevent leakage: DepTime, DepDelayMinutes, DepDel15, ArrTime, ActualElapsedTime, AirTime, ArrDelayMinutes, ArrDel15 (except as target source), and all delay-cause fields (CarrierDelay, WeatherDelay, NASDelay, SecurityDelay, LateAircraftDelay).

2.3 Validation Strategy

Time-aware validation was used throughout to avoid look-ahead bias:

- **Rolling cross-validation** on January–September: train through month N, validate on month N+1, repeated for July/August/September folds.
- **Final holdout** on October–December, evaluated only once after model and threshold selection.

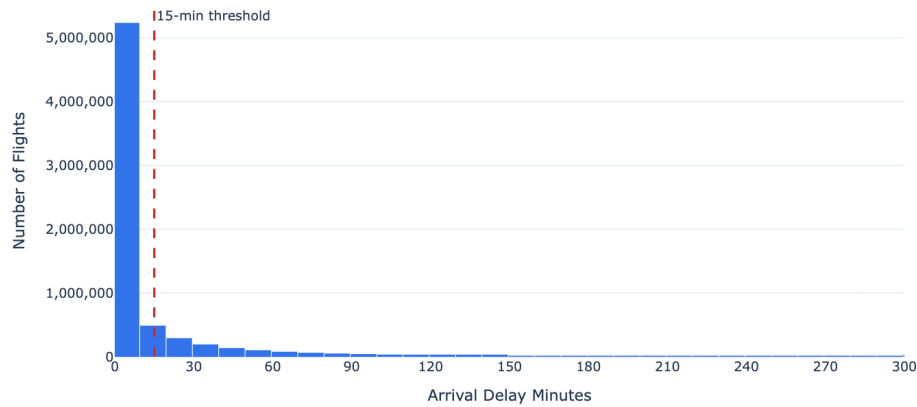
This setup mimics realistic deployment: a model trained on historical data must generalize to future flights, not flights from random months mixed across the calendar.

3. EDA Findings

3.1 Overall Delay Distribution

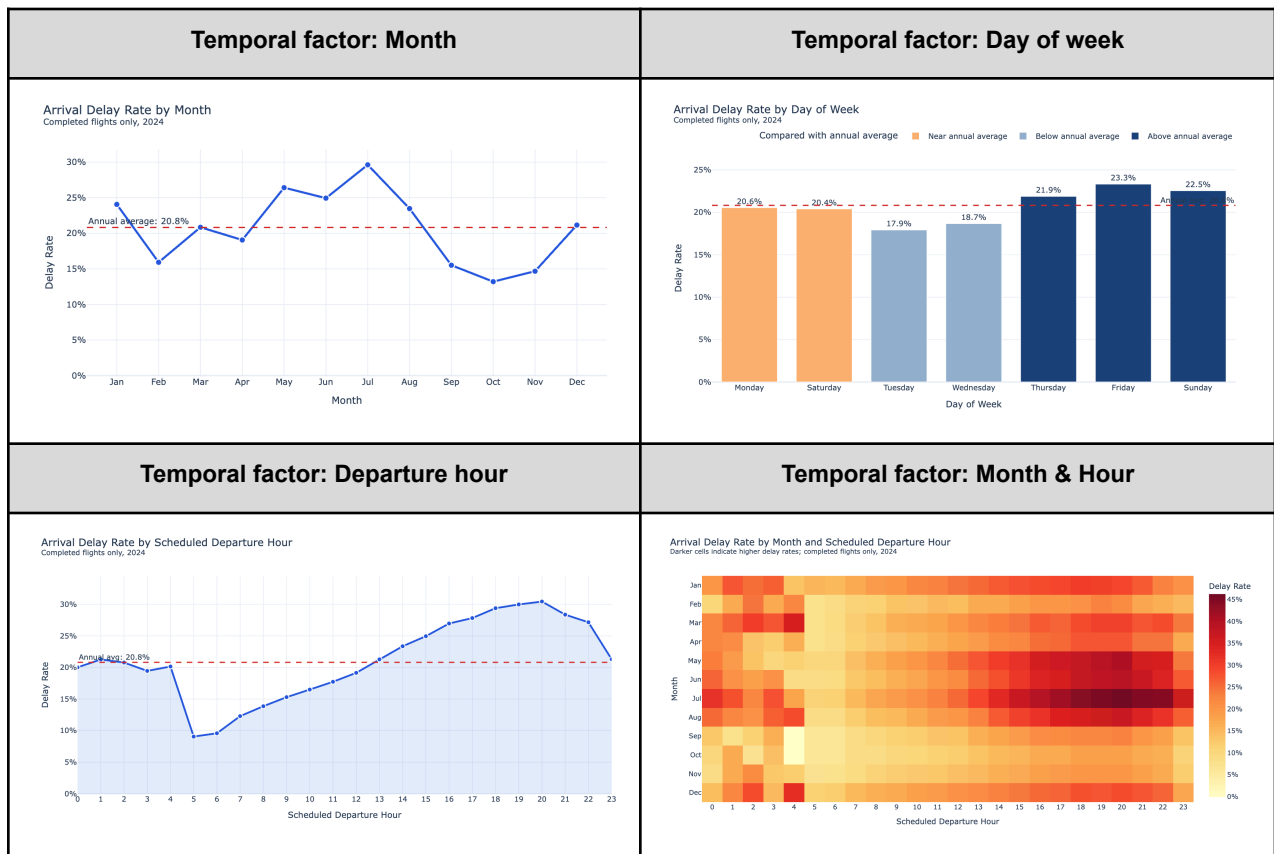
Arrival delay minutes are highly right-skewed. The median completed flight has 0 minutes of delay, but the 95th percentile is 83 minutes and extreme delays exceed 3,000 minutes, likely reflecting severe operational disruptions. Using the binary 15+ minute threshold, 20.8% of completed flights were delayed in 2024.

Distribution of Arrival Delay Minutes
Values above 300 minutes are capped for readability



3.2 Temporal Patterns

Time is one of the clearest signals in the analysis. Delay rates are lowest in the early morning and rise through the afternoon and evening, peaking around 19:00-20:00 scheduled departure time. This pattern suggests that delays accumulate as the operating day progresses.



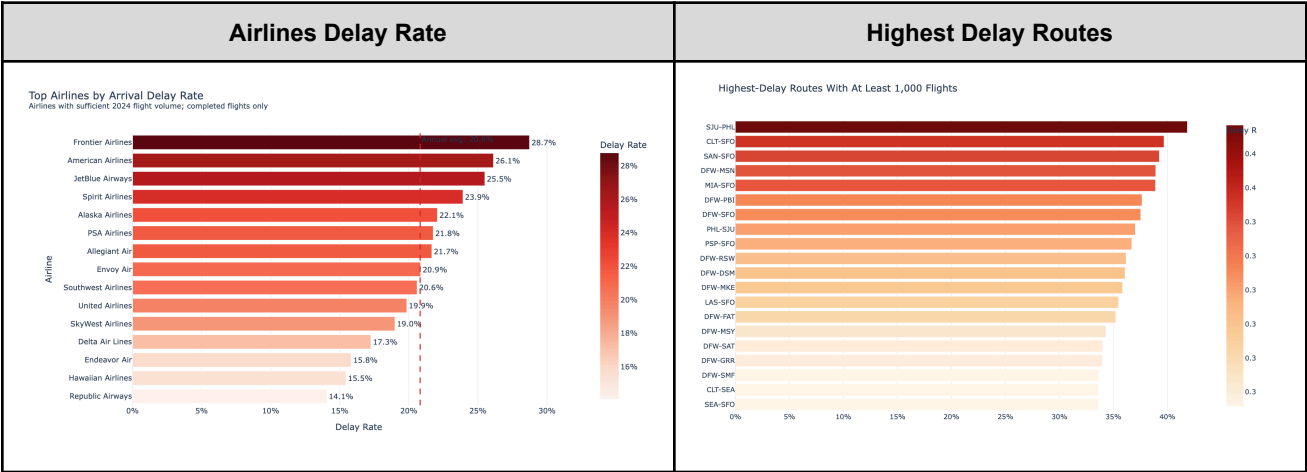
Temporal factor	Finding	Modeling implication
Month	July had the highest monthly delay rate at about 30%; October and November were among the lowest.	Include seasonality and use time-aware validation.
Day of week	Friday and Sunday had higher delay rates than Tuesday and Wednesday.	Day-of-week effects are useful, but smaller than hour effects.
Departure hour	Delay risk rose from early morning into evening, reaching about 30% around 19:00-20:00.	Scheduled departure hour is a key predictor.
Weekend flag	Weekend delay rate was slightly higher than weekday delay rate.	Weekend flag is useful but not a dominant driver.

3.3 Airline, Airport, and Route Patterns

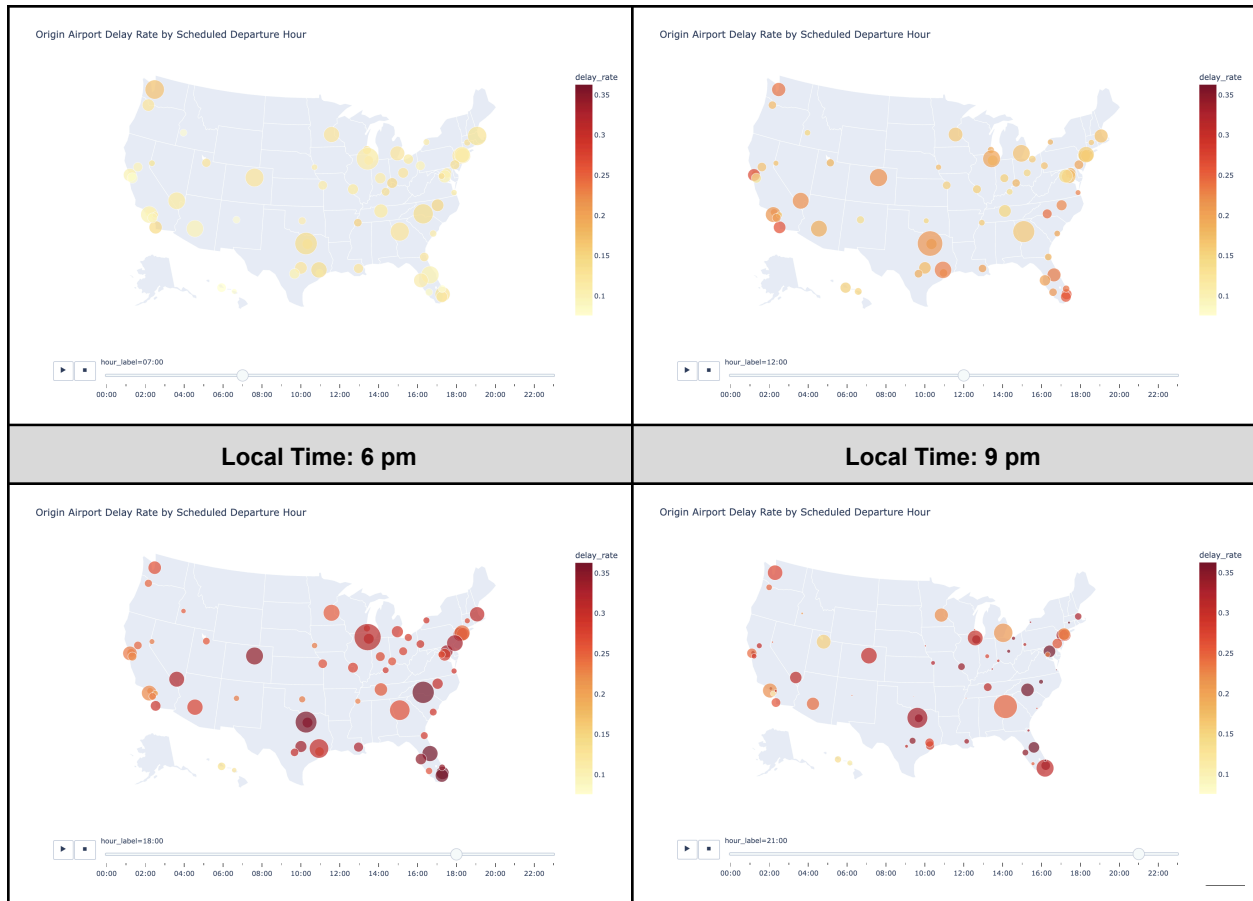
Delay rates varied substantially by carrier, airport, and route.:

- **Carriers:** Frontier (29%), American (26%), JetBlue (26%) at the top; Hawaiian (15%) and Republic (14%) at the bottom.
- **Origin airports:** Miami, DFW, Charlotte, and Fort Lauderdale all exceed 27% — hub airports and Florida airports stand out.
- **Routes:** SJU–PHL leads at 42%, with multiple DFW and SFO routes exceeding 35%.

These patterns support including carrier, origin airport, destination airport, and route features in the model. At the same time, airport and route effects should not be interpreted causally, because they may proxy for congestion, geography, carrier network structure, weather exposure, and route scheduling patterns.



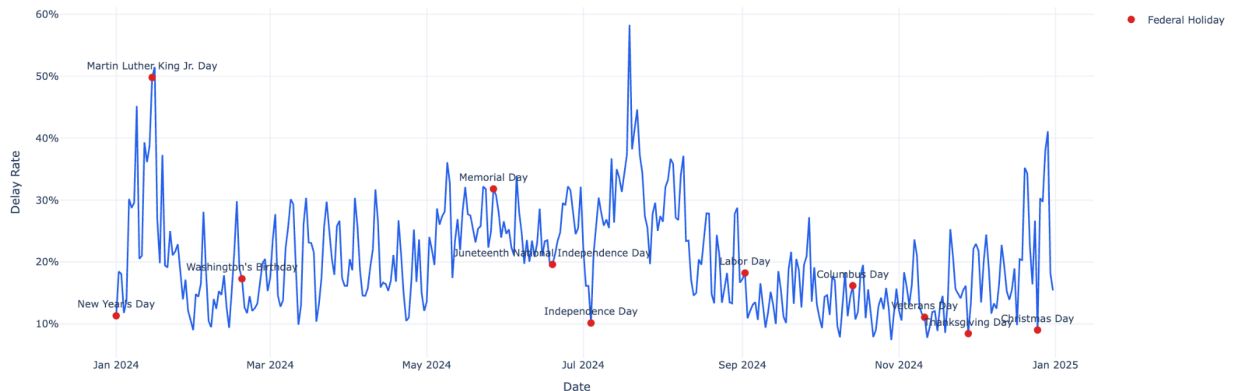
Delay Rate by airport & time of the day	
Local Time: 7 am	Local Time: 12 pm



3.4 Holiday Effects

Federal holidays show high variance, not a uniform pattern. Martin Luther King Jr. Day reached a 50% delay rate in 2024, while Thanksgiving Day (8.5%) and Christmas Day (9.0%) were below average. The `is_holiday_week` (± 2 days) feature captures this better than `is_holiday` alone because travel volume often shifts to days adjacent to the holiday.

Daily Arrival Delay Rate with Federal Holidays
Completed flights only, 2024



4. Modeling Results

4.1 Model Comparison

Three candidates were evaluated using rolling cross-validation on January–September:

Model	ROC-AUC	Precision	Recall	F1
Logistic regression (t=0.40)	0.65	0.26	0.90	0.40
XGBoost (t=0.20)	0.67	0.31	0.72	0.42

XGBoost provides the best overall tradeoff across ranking quality, F1, and operating point. Logistic regression's recall is higher at the same operating point only because it flags a larger share of flights, not because it ranks them better.

4.2 Threshold Tuning

The default 0.50 threshold is not a good choice for this problem. Because only ~21% of flights are delayed, the model's probability distribution is concentrated below 0.50, and the default threshold flags almost nothing. The cross-validation grid revealed:

- **At threshold 0.50:** precision=0.33, recall=0.01 — the model almost never predicts delay.
- **At threshold 0.20:** precision=0.31, recall=0.72 — catches most delays at a manageable false positive rate.

Threshold 0.20 was selected as the operating point based on cross-validation F1, with the understanding that recall is the more important metric for a delay-screening use case.

4.3 Final Holdout Performance

Because flight delay patterns vary over time, the model uses rolling monthly cross-validation instead of random K-fold validation. The validation folds train on earlier months and validate on later months. October–December is reserved as a final future holdout period.

Refitting XGBoost on all of January–September and evaluating on October–December:

Metric	Cross-Validation	Holdout (Oct–Dec)
ROC-AUC	0.67	0.62
Precision	0.31	0.21
Recall	0.72	0.67
F1	0.42	0.32
Flagged share	—	53%

The drop in precision from 0.31 to 0.21 reflects a real shift in conditions: the October–December delay rate is 16.3% vs 23%+ during the CV folds. Because the base rate is lower, the same absolute false positive count produces a worse precision figure. ROC-AUC dropped less (0.67 to 0.62), suggesting the model's underlying ranking ability is preserved — it's the threshold calibration that suffered from distribution shift.

At the selected 0.20 threshold, the model flags 53% of holdout flights as delay-risk flights. This is reasonable for a broad screening or ranking use case where missed delays are costly, but it would need a higher threshold or additional real-time features for applications where false alarms carry a meaningful cost.

4.4 Feature Importance

Permutation importance was used to interpret the final XGBoost model. This method measures how much holdout ROC-AUC drops when each feature is randomly shuffled. Importance was displayed as percentage-point ROC-AUC drop for readability.

The most important features align with the EDA: delay risk depends strongly on time of day, carrier operations, airport/network structure, and weekly patterns.

Rank	Feature	Interpretation
1	Scheduled departure hour	Most important predictor; captures accumulation of delays across the operating day.
2	Reporting airline	Carrier-level operational differences and network structure affect delay risk.
3	Scheduled arrival hour	Captures additional timing patterns and correlation with departure hour.
4	Day of week	Weekly travel and operating patterns affect delay risk.
5	Destination airport	Airport and network effects contribute to delay risk.

5. Insights and Business Implications

5.1 What Predicts Delays

Hour of departure, month, and airline are the strongest pre-flight signals. The model effectively encodes:

- **Cascading delay risk:** evening flights inherit delay from earlier flights using the same aircraft and crew, even though the late-aircraft delay field is excluded from the model. This is what makes hour-of-day such a strong predictor: it implicitly captures the cascading-delay structure of the operating day, even though we never gave the model that field directly.
- **Seasonal weather risk:** July's spike is consistent with summer weather and operational disruption patterns, though weather data would be needed to confirm the cause directly.

- **Carrier operational efficiency:** legacy carriers and ultra-low-cost carriers show different baseline delay rates, likely reflecting differences in turnaround buffers and route concentration.

5.2 What This Could Be Used For

- **Passenger-facing delay alerts:** at recall=0.67, the model catches two-thirds of delays — a reasonable early-warning system if combined with day-of-departure signals.
- **Airline scheduling:** the high-risk hour-airport-route combinations identified here could inform where airlines need additional buffer time in their schedules.
- **Airport operations planning:** the airports in the top decile of delay rate are candidates for capacity or scheduling intervention.

5.3 Limitations

The model is a baseline, not a finished product. Specific gaps:

- **Data volume sampling.** The model uses a 500,000-row stratified sample for computational feasibility in Colab. A production version should be trained and validated on the full dataset or a larger sample.
- **No weather data.** Weather is likely one of the most important external delay drivers not currently in the model. Adding hourly weather at origin and destination is the potential highest-ROI improvement.
- **No cascading-delay proxy.** Historical delay rate of the specific aircraft tail number over the prior 24 hours would be highly predictive but was not included due to feature-engineering complexity.
- **No congestion features.** Same-airport same-hour scheduled departure volume was scoped but not run due to memory constraints in this environment.
- **Distribution shift not addressed.** The precision drop from CV to holdout suggests the model would benefit from regular recalibration or a more adaptive threshold approach in production.

6. Future Improvements

Improvements to the model can be considered:

- **Add weather features.** Hourly precipitation, wind speed, visibility, and gust at origin and destination airports. Expected to improve performance because weather is a major external driver of flight delays.
- **Add cascading delay features.** If available, aircraft tail-number rotation features could capture cascading delay effects by linking each flight to the delay status of the aircraft's prior flight.

- **Cross-year out-of-sample validation.** The EDA revealed substantially different delay patterns across months — Q3 is driven by summer thunderstorms, Q4 may reflect a different seasonal regime from summer, including holiday travel patterns and early winter weather exposure. The current Oct–Dec 2024 holdout, while a valid out-of-time test, covers only one seasonal regime. A more rigorous validation would train on 2024 full year and evaluate on stratified monthly samples from 2025 (e.g., 30k flights per month, balanced across hours). This would address two limitations the current setup cannot: distribution shift across years (carrier operations, fleet changes), and seasonal generalization across all twelve months rather than just Q4. In production, this would become continuous monitoring — each new month becomes an additional out-of-sample evaluation point, with retraining triggered when monthly AUC falls below threshold.
- **Add origin-airport congestion.** Number of scheduled departures at the origin airport in the same hour. Captures peak-bank effects that hour-of-day alone misses.
- **Time-aware target encoding.** Replace one-hot encoding of high-cardinality fields (route, origin, dest) with smoothed historical delay rates computed using only data prior to each flight — strictly avoiding look-ahead bias.
- **Build a separate post-departure model.** Once a flight has departed, DepDelayMinutes becomes available and is highly predictive of arrival delay (Spearman correlation 0.82). A post-departure model would have much higher accuracy and would serve a different operational use case (mid-flight rebooking, gate planning).
- **Production calibration monitoring.** Track precision and recall by week in production and trigger threshold recalibration when distribution shifts.

7. Conclusion

The analysis shows that arrival delay risk is systematic but only partially predictable from pre-flight schedule and network data. Delay risk varies meaningfully by time of day, season, carrier, airport, and route, and XGBoost provided the strongest cross-validated screening model after threshold tuning.

However, holdout performance confirms that flight delays are also driven by operational factors not fully captured in this dataset. The model is therefore best positioned as an early risk-ranking tool rather than a final delay classifier. At the selected threshold, it favors recall and broad screening, which is useful when missing a potential delay is more costly than flagging extra flights.

Future iterations should add weather, aircraft rotation, airport congestion, crew/equipment availability, and live operational signals to improve precision and generalization. I would also validate the model on future months, recalibrate probability thresholds by season or airport, and monitor performance drift before using it in production.